

A-level

# ENVIRONMENTAL SCIENCE

## Paper 1 - Generated practice paper Set 26

### Mark scheme

120 marks: Questions 1-10 (95 marks) plus one essay from Question 11 (25 marks).

### Marking guidance

- The number in brackets after each marking point is the mark it earns; the note above the points says how many are needed.
- Where a question asks for a fixed number of points (eg *two reasons*), credit only the first points given.
- Alternative correct wording and other relevant, accurate points are credited.
- For levels-of-response questions, first determine the level using the descriptors, then position the mark within the level using the indicative content.
- Calculations: credit correct working (method marks) even if the final answer is wrong; incorrect answers with no working score 0.

**Textbook references:** every question below cites the printed page numbers in R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018) from which the content is drawn, together with the AQA 7447 specification reference.

| Question | Marking guidance                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Marks |
|----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| 01.1     | <i>1 mark for each point.</i> <ul style="list-style-type: none"> <li>• Spearman's rank correlation coefficient (1)</li> <li>• Chi-squared test (frequencies / counts in categories) (1)</li> <li>• t-test (two means of measured data) (1)</li> <li>• Spearman's rank correlation coefficient (1)</li> </ul>                                                                                                                                                         | 4     |
| 01.2     | <i>1 mark.</i> <ul style="list-style-type: none"> <li>• Chi-squared compares frequencies / counts in categories; it cannot be used for continuously variable measured data (1)</li> </ul>                                                                                                                                                                                                                                                                            | 1     |
| 01.3     | <i>1 mark for each point.</i> <ul style="list-style-type: none"> <li>• The value (ignoring sign) exceeds the critical value so the correlation is significant at the 5% level (1)</li> <li>• The negative sign shows a negative correlation: one variable decreases as the other increases (1)</li> <li>• There is less than a 5% probability the correlation arose by chance; reject the null hypothesis (but correlation is not proof of causation) (1)</li> </ul> | 3     |
| 01.4     | <i>1 mark for each point.</i> <ul style="list-style-type: none"> <li>• Show the standard deviation / range of the original values around each mean (1)</li> <li>• If the bars of two means overlap greatly the difference is probably not significant; little overlap suggests a significant difference (1)</li> </ul>                                                                                                                                               | 2     |
|          | <b>Source:</b> R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 422-423; AQA spec 3.7.2.3.                                                                                                                                                                                                                                                                                                                                              |       |

| Question    | Marking guidance                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Marks |
|-------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| <b>02.1</b> | <p><i>1 mark for each point.</i></p> <ul style="list-style-type: none"> <li>• Visible light passes through the atmosphere easily and is absorbed by the Earth's surface, which warms and emits infrared <b>(1)</b></li> <li>• Greenhouse gases (naturally water vapour and CO<sub>2</sub>) absorb infrared better than the average for the atmosphere, converting it to heat and re-emitting some back to the surface <b>(1)</b></li> <li>• This keeps the surface about 33 °C warmer than it would otherwise be <b>(1)</b></li> </ul> | 3     |
| <b>02.2</b> | <p><i>1 mark for each point.</i></p> <ul style="list-style-type: none"> <li>• CO<sub>2</sub> is emitted in far larger quantities (every combustion process, deforestation) so its total effect is greatest <b>(1)</b></li> <li>• CO<sub>2</sub> has a long residence time so it accumulates; methane has a short residence time (12 years) so it is removed faster <b>(1)</b></li> <li>• CFC emissions are now very small following the Montreal Protocol <b>(1)</b></li> </ul>                                                        | 3     |
| <b>02.3</b> | <p><i>1 mark for each point.</i></p> <ul style="list-style-type: none"> <li>• Methane has a much higher global warming potential (28x) so each tonne removed has a bigger effect <b>(1)</b></li> <li>• Its short residence time (about 12 years) means atmospheric concentrations fall quickly once emissions are cut <b>(1)</b></li> </ul>                                                                                                                                                                                            | 2     |
| <b>02.4</b> | <p><i>Any 2 from the points below - 1 mark each, maximum 2.</i></p> <ul style="list-style-type: none"> <li>• Collect landfill / anaerobic digester gas and use as fuel <b>(1)</b></li> <li>• Changes in cattle diet / fewer cattle / less meat consumption <b>(1)</b></li> <li>• Draining rice paddies periodically <b>(1)</b></li> <li>• Anaerobic digestion of manure with gas capture <b>(1)</b></li> </ul>                                                                                                                         | 2     |
|             | <b>Source:</b> R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 89-90; AQA spec 3.2.1.2.                                                                                                                                                                                                                                                                                                                                                                                                                  |       |
| <b>03.1</b> | <p><i>1 mark for the correct answer.</i></p> <ul style="list-style-type: none"> <li>• <math>(120 + 95) - (2 + 45) = 168 \text{ t ha}^{-1}</math> <b>(1)</b></li> </ul>                                                                                                                                                                                                                                                                                                                                                                 | 1     |
| <b>03.2</b> | <p><i>1 mark for each point.</i></p> <ul style="list-style-type: none"> <li>• Annual ploughing aerates the soil so aerobic decomposers break down dead organic matter faster, releasing CO<sub>2</sub> <b>(1)</b></li> <li>• Harvest removes biomass so less dead organic matter is returned; pasture roots and litter constantly add organic matter <b>(1)</b></li> </ul>                                                                                                                                                             | 2     |
| <b>03.3</b> | <p><i>Any 2 from the points below - 1 mark each, maximum 2.</i></p> <ul style="list-style-type: none"> <li>• Low / zero tillage <b>(1)</b></li> <li>• Adding manure, compost or mulch <b>(1)</b></li> <li>• Growing cover crops / leaving crop residues <b>(1)</b></li> <li>• Converting to long-term grass or tree crops in rotation <b>(1)</b></li> </ul>                                                                                                                                                                            | 2     |
|             | <b>Source:</b> R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 148-151; AQA spec 3.2.4.2.                                                                                                                                                                                                                                                                                                                                                                                                                |       |
| <b>04.1</b> | <p><i>Correct answer only.</i></p> <ul style="list-style-type: none"> <li>• A waste is assumed to cause pollution until research shows it is unlikely to do so <b>(1)</b></li> </ul>                                                                                                                                                                                                                                                                                                                                                   | 1     |
| <b>04.2</b> | <p><i>1 mark for each point.</i></p> <ul style="list-style-type: none"> <li>• If the cost of preventing damage is less than the (penalty for) damage there is a financial incentive to prevent pollution <b>(1)</b></li> <li>• If polluters can escape their responsibilities / penalties are low, there is little incentive to comply <b>(1)</b></li> </ul>                                                                                                                                                                           | 2     |

| Question    | Marking guidance                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Marks |
|-------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| <b>04.3</b> | <p><i>Any 2 from the points below - 1 mark each, maximum 2.</i></p> <ul style="list-style-type: none"> <li>• Marine discharges where currents dilute and disperse emissions <b>(1)</b></li> <li>• Emissions downwind of urban areas <b>(1)</b></li> <li>• Not discharging waste onto permeable rock above an aquifer <b>(1)</b></li> </ul>                                                                                                                                                                                                                                                                             | 2     |
|             | <b>Source:</b> R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 243-244; AQA spec 3.4.3.1.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |       |
| <b>05.1</b> | <p><i>1 mark for each point.</i></p> <ul style="list-style-type: none"> <li>• Booms are floating tubes with a skirt that restrict the movement of floating oil, keeping it near the source / away from sensitive sites (mangroves, reef, marine park) <b>(1)</b></li> <li>• They work best in sheltered / calm water such as the lagoon <b>(1)</b></li> <li>• Limitation: they do not work in open water where currents or waves are strong; oil can flow under the skirt in a strong current <b>(1)</b></li> </ul>                                                                                                    | 3     |
| <b>05.2</b> | <p><i>1 mark for each point.</i></p> <ul style="list-style-type: none"> <li>• Skimmers: rotating (metal) discs / belts pick up floating oil which is scraped off and stored <b>(1)</b></li> <li>• Absorbent materials (high surface area, eg textile mops, straw, human hair) soak up oil and are then removed <b>(1)</b></li> <li>• Collected oil / absorbents are incinerated or sent to landfill (or oil recycled) <b>(1)</b></li> </ul>                                                                                                                                                                            | 3     |
| <b>05.3</b> | <p><i>1 mark for each point.</i></p> <ul style="list-style-type: none"> <li>• Dispersants break the oil into an emulsion of droplets, making it more mobile so it spreads through the water column (to the reef / seabed organisms) rather than staying on the surface <b>(1)</b></li> <li>• They do not reduce the amount of oil; the dispersant / oil mixture may be toxic to corals and plankton in shallow, enclosed water <b>(1)</b></li> </ul>                                                                                                                                                                   | 2     |
| <b>05.4</b> | <p><i>1 mark for each point.</i></p> <ul style="list-style-type: none"> <li>• Bacteria that break down hydrocarbons digest the residual oil (most rapid where temperature, oxygen and nutrients are high) <b>(1)</b></li> <li>• Accelerated by adding nutrients (nitrates / phosphates) and aerating (ploughing) / irrigating the sediment <b>(1)</b></li> </ul>                                                                                                                                                                                                                                                       | 2     |
|             | <b>Source:</b> R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 261-262; AQA spec 3.4.3.2.7.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |       |
| <b>06.1</b> | <p><i>1 mark for the correct answer.</i></p> <ul style="list-style-type: none"> <li>• <math>15 - (-18) = 33\text{ }^{\circ}\text{C}</math> <b>(1)</b></li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 1     |
| <b>06.2</b> | <p><i>1 mark for each point.</i></p> <ul style="list-style-type: none"> <li>• Visible light passes easily through the atmosphere and is absorbed by the surface, which warms <b>(1)</b></li> <li>• The warm surface emits (long-wave / far) infrared radiation <b>(1)</b></li> <li>• Greenhouse gases (water vapour, <math>\text{CO}_2</math>, methane) absorb infrared better than the average for the atmosphere and convert it to heat <b>(1)</b></li> <li>• The warm atmosphere emits infrared which is absorbed by the surface, and it reduces heat loss from land and oceans by conduction <b>(1)</b></li> </ul> | 4     |
| <b>06.3</b> | <p><i>1 mark.</i></p> <ul style="list-style-type: none"> <li>• Any gas that is better at absorbing infrared radiation than the average for the atmosphere <b>(1)</b></li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                        | 1     |

| Question    | Marking guidance                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Marks |
|-------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| <b>06.4</b> | <p>1 mark for each point.</p> <ul style="list-style-type: none"> <li>Arriving: mainly ultraviolet, visible light and near infrared (short wavelength); leaving: mainly long-wavelength (far) infrared (1)</li> <li>Gases that are transparent to the incoming short wavelengths can absorb the outgoing long wavelengths, so changing their concentration changes how much energy is retained (1)</li> </ul>                                                                                                                                            | 2     |
| <b>06.5</b> | <p>1 mark for each point.</p> <ul style="list-style-type: none"> <li>The energy and the processes it drives control climate, ocean currents and the hydrological cycle (1)</li> <li>These determine the abiotic conditions (temperature, rainfall) that each species is adapted to, so a change alters where species can survive (1)</li> </ul>                                                                                                                                                                                                         | 2     |
|             | <b>Source:</b> R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 87-89; AQA spec 3.2.1.1.                                                                                                                                                                                                                                                                                                                                                                                                                                   |       |
| <b>07.1</b> | <p>A correct final answer (with unit where asked) scores full marks. If the answer is wrong, award 1 mark for each correct step of working shown.</p> <ul style="list-style-type: none"> <li><math>13 / 480 = 0.0271</math> years (1)</li> <li><math>0.0271 \times 365</math> (1)</li> <li><math>= 10</math> days (accept 9.9-10) (1)</li> </ul>                                                                                                                                                                                                        | 3     |
| <b>07.2</b> | <p>1 mark for the correct answer.</p> <ul style="list-style-type: none"> <li><math>1\,340\,000 / 425 = 3153</math> years (accept 3100-3200) (1)</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                              | 1     |
| <b>07.3</b> | <p>1 mark for each point.</p> <ul style="list-style-type: none"> <li>It has a very short residence time / very rapid transfer rate (<math>480 \times 10^{15}</math> dm<sup>3</sup> per year) (1)</li> <li>So it moves large volumes of water each year, delivering precipitation to land; small reservoirs with fast transfers respond quickly to change (1)</li> </ul>                                                                                                                                                                                 | 2     |
| <b>07.4</b> | <p>1 mark for each point.</p> <ul style="list-style-type: none"> <li>The ocean is so large that an added or removed volume is a tiny proportion of the total (1)</li> <li>The same volume is a large proportion of a small reservoir so its level / residence time changes noticeably (1)</li> </ul>                                                                                                                                                                                                                                                    | 2     |
| <b>07.5</b> | <p>1 mark for each point.</p> <ul style="list-style-type: none"> <li>Water is constantly moving between reservoirs (evaporation, precipitation, runoff, infiltration, transpiration) (1)</li> <li>But inflows and outflows for each reservoir balance so the volume in each reservoir stays constant over long periods (1)</li> </ul>                                                                                                                                                                                                                   | 2     |
|             | <b>Source:</b> R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 115-116; AQA spec 3.2.2.1.                                                                                                                                                                                                                                                                                                                                                                                                                                 |       |
| <b>08.1</b> | <p>2 marks for each point (partial credit for a partly correct point).</p> <ul style="list-style-type: none"> <li>Overgrazing: vegetation is eaten faster than it grows so soil is exposed; hooves disturb the soil, damage roots and compact it, reducing infiltration and increasing runoff (2)</li> <li>Reduced biota (ploughing, agrochemicals, less organic matter): less humus so less adhesion between particles; fewer worms so less aeration and lower infiltration capacity; fewer nutrients released so less vegetation cover (2)</li> </ul> | 4     |
| <b>08.2</b> | <p>1 mark for each point.</p> <ul style="list-style-type: none"> <li>Tree roots that held the soil together decompose so the soil is bound less strongly (1)</li> <li>Heavy rain adds mass and lubricates particles; without interception more water enters the soil, making slumping more likely (1)</li> </ul>                                                                                                                                                                                                                                        | 2     |

| Question    | Marking guidance                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Marks |
|-------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| <b>08.3</b> | <p><i>Any 4 from the points below - 1 mark each, maximum 4.</i></p> <ul style="list-style-type: none"> <li>• Sedimentation in rivers reduces flow capacity so flooding downstream is more likely <b>(1)</b></li> <li>• Sedimentation in reservoirs reduces their storage volume <b>(1)</b></li> <li>• Coastal sedimentation covering reefs / seagrass; turbidity reducing light for aquatic plants <b>(1)</b></li> <li>• Increased atmospheric particulates from wind erosion worsening asthma; dust covering crops / urban areas <b>(1)</b></li> <li>• Desertification as vegetation cannot re-establish; reduced rainfall <b>(1)</b></li> <li>• Nutrients (phosphates) carried on soil particles causing eutrophication <b>(1)</b></li> </ul> | 4     |
|             | <b>Source:</b> R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 162-164; AQA spec 3.2.5.2.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |       |
| <b>09.1</b> | <p><i>1 mark for each point.</i></p> <ul style="list-style-type: none"> <li>• Mutagen: an agent that changes the chemical structure / base sequence of DNA (causes mutations), eg ionising radiation, UV, PCBs, cadmium, asbestos <b>(1)</b></li> <li>• Carcinogen: a mutagen that causes cancer / uncontrolled cell division producing a tumour <b>(1)</b></li> <li>• Teratogen: an agent that causes birth abnormalities by preventing normal gene expression (inhibiting proteins / enzymes) without changing DNA structure, eg mercury, 2,4,5-T <b>(1)</b></li> </ul>                                                                                                                                                                       | 3     |
| <b>09.2</b> | <p><i>1 mark for each point.</i></p> <ul style="list-style-type: none"> <li>• Teratogens do not change the DNA structure, so the abnormality is not in the genes and cannot be passed on <b>(1)</b></li> <li>• Mutagens change DNA; if the mutation is in a gamete / egg / sperm (gonadic effect) it can be inherited <b>(1)</b></li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                     | 2     |
| <b>09.3</b> | <p><i>1 mark for each point.</i></p> <ul style="list-style-type: none"> <li>• Somatic: mutation in a general body cell, affecting the individual only (eg cancer / tumour) <b>(1)</b></li> <li>• Gonadic: mutation in cells of the ovaries or testes / gametes / embryo, which may cause abnormalities in offspring <b>(1)</b></li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                       | 2     |
| <b>09.4</b> | <p><i>1 mark for each point.</i></p> <ul style="list-style-type: none"> <li>• Two or more pollutants whose effects interact to produce a different (usually more serious) effect <b>(1)</b></li> <li>• The pollutants themselves do not react to form a new pollutant (that would be a secondary pollutant) <b>(1)</b></li> <li>• Example: ozone damages leaf cuticles so sulfur dioxide can cause more damage to the exposed cells; smoke and sulfur dioxide in smogs; neonicotinoids with fungicides <b>(1)</b></li> </ul>                                                                                                                                                                                                                    | 3     |
|             | <b>Source:</b> R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 239-240; AQA spec 3.4.1.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |       |
| <b>10.1</b> | <p><i>1 mark for the correct answer.</i></p> <ul style="list-style-type: none"> <li>• <math>4.2 / 0.7 = 6</math> <b>(1)</b></li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 1     |
| <b>10.2</b> | <p><i>1 mark for each point.</i></p> <ul style="list-style-type: none"> <li>• Long summer days (almost 24 hours of daylight) so the total energy received is high, even though the Sun is low <b>(1)</b></li> <li>• In winter days are very short / the Sun does not rise, so light levels are extremely low <b>(1)</b></li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                              | 2     |

| Question | Marking guidance                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Marks |
|----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| 10.3     | <p>1 mark for each point.</p> <ul style="list-style-type: none"> <li>• Reflectors can only focus parallel rays of direct sunlight onto the absorber (1)</li> <li>• Clouds, smoke and dust scatter the light so it cannot be focused and intensified (1)</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | 2     |
| 10.4     | <p>1 mark.</p> <ul style="list-style-type: none"> <li>• Very large areas of collectors are needed to harness significant amounts of energy (and the changing angle of incidence reduces it further) (1)</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | 1     |
| 10.5     | <p><b>Levels of response</b></p> <p><b>Level 3 (7-9 marks):</b> Answer is comprehensive, relevant and logically structured. Shows detailed knowledge and understanding of the relevant scientific ideas and applies them accurately to the context. Evaluation/analysis is well supported by evidence and a clear judgement or conclusion is reached.</p> <p><b>Level 2 (4-6 marks):</b> Answer is mostly relevant and shows some structure. Knowledge and understanding of relevant ideas is sound but application to the context or evaluation may be incomplete or lack detail. A judgement may be made but is not fully supported.</p> <p><b>Level 1 (1-3 marks):</b> Answer is basic, with limited structure. Some relevant knowledge is shown but there are omissions or errors. Little application to the context and little or no evaluation.</p> <p><b>0 marks:</b> No relevant content.</p> <p><b>Indicative content</b> (credit other relevant, accurate points)</p> <ul style="list-style-type: none"> <li>• Problems: intermittency (daily and seasonal cycles), unpredictable cloud cover, low energy density, changing angle of incidence, land requirements, water for cleaning in deserts, toxic wastes in manufacture (silicon tetrachloride, cadmium), efficiency loss at high temperature</li> <li>• Photothermal: evacuated tubes reduce conduction / convection losses; thermal stores (water, sand, concrete, molten salt) allow later use; passive solar architecture, brise soleil, light tubes; heat pumps deliver up to four times the energy used</li> <li>• Photovoltaic developments: multijunction cells absorbing more wavelengths, anti-reflective / moth-eye textured surfaces (smooth cells reflect 30%), PV-thermal hybrids cooling the cell, transparent PV in windows, heliostats keeping the optimum angle, self-cleaning nanohydrophobic surfaces</li> <li>• Concentrating solar power with molten-salt storage to 550 °C so steam turbines run when needed - overcomes intermittency</li> <li>• Integration: grid-connected solar farms, rooftop space avoiding land conflicts, batteries, pumped storage, links to energy storage and demand management</li> <li>• Evaluation: strong in sunny deserts / low latitudes, weaker in high latitudes in winter; cost falling; manufacturing impacts; no single technology solves all problems; justified conclusion</li> </ul> | 9     |
|          | <b>Model answer - Level 3 (9 marks)</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |       |
|          | <p>Solar power's problems are intrinsic: it is intermittent with daily and seasonal cycles, cloud makes it unpredictable and scatters light so it cannot be focused, its low energy density demands large collector areas, the angle of incidence keeps changing, cleaning in deserts uses scarce water, manufacturing PV produces toxic wastes such as silicon tetrachloride and cadmium, and PV efficiency falls as cells heat up.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |       |

| Question | Marking guidance                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Marks |
|----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
|          | <p>New technology addresses most of these. Evacuated-tube photothermal panels cut conduction and convection losses, and thermal stores of water, sand, concrete or molten salt hold heat for later; passive solar design, brise soleil and light tubes reduce demand; heat pumps deliver up to four times the energy used to run them. Multijunction PV cells absorb more wavelengths, moth-eye anti-reflective surfaces stop the 30% of light a smooth cell reflects, PV-thermal hybrids cool the cell and use the heat, transparent PV turns windows into generators, heliostats keep panels at the optimum angle and nanohydrophobic surfaces self-clean.</p> <p>Concentrating solar power with molten salt stored at 550 °C runs steam turbines when electricity is wanted, directly overcoming intermittency, and grid-connected farms, rooftops that avoid land conflicts, batteries and pumped storage integrate the output.</p> |       |
|          | <p>Even so, no technology changes the fact that winter insolation at high latitudes is tiny, that CSP needs cloudless skies, or that manufacturing has impacts. Solar can dominate supply in sunny, low-latitude regions and contribute strongly elsewhere in summer, but overcoming its problems fully depends on storage and a diverse energy mix rather than on solar technology alone.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |       |
|          | <b>Weaker answer - Level 1 (2 marks)</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |       |
|          | <p>Solar power has problems because it does not work at night or when it is cloudy, and solar panels are not very efficient. New technologies can help. Better solar panels are more efficient. Solar power can be stored in batteries for use at night. Mirrors can concentrate the sunlight to make it hotter. So new technologies can overcome most of the problems with solar power.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |       |
|          | <b>What makes the difference</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |       |
|          | <ul style="list-style-type: none"> <li>• Level 1 identifies two problems and three vague technologies; Level 3 names each technology and explains which problem it solves and how.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |       |
|          | <ul style="list-style-type: none"> <li>• Level 3 acknowledges what technology cannot fix (high-latitude winters, cloud for CSP, manufacturing impacts) - the 'extent' in the question.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |       |
|          | <ul style="list-style-type: none"> <li>• Numbers (30% reflection, 550 °C, four-fold heat-pump output) show detailed knowledge applied to the argument.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |       |
|          | <p><b>Source:</b> R. Genn, Environmental Science A level AQA (Insight &amp; Perspective, 2018), pp. 199-203; AQA spec 3.3.4.1.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |       |

| Question | Marking guidance                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Marks |
|----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| 11.1     | <p><b>Evaluate the contribution of new technologies to the monitoring and understanding of environmental change in the physical environment.</b></p> <p><b>Levels of response</b></p> <p><b>Level 5 (21-25 marks):</b> Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached.</p> <p><b>Level 4 (16-20 marks):</b> Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified.</p> <p><b>Level 3 (11-15 marks):</b> Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics.</p> <p><b>Level 2 (6-10 marks):</b> Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions.</p> <p><b>Level 1 (1-5 marks):</b> Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate.</p> <p><b>0 marks:</b> No relevant content.</p> <p><b>Indicative content</b> (students are not expected to cover all of these; credit other relevant material)</p> <ul style="list-style-type: none"> <li>• Historical data limitations; proxy data - ice cores, isotope ratios (O-18:O-16, C-14), Mg:Ca in calcite, tree rings, pollen, coral</li> <li>• Satellite sensors: visible light imagery, infrared for vegetation and temperature, radar microwaves for wind, wave height, sea level and oil, weather monitoring; passing through cloud</li> <li>• Sensor carriers: ROVs / drones, AUVs, balloons, aircraft, animals; data recorders for abiotic factors; GPS tracking of floats and sampling sites; Argo floats and ocean monitoring</li> <li>• Climate models and their assumptions; monitoring atmospheric composition (Mauna Loa), ozone (Dobson units), sea level and ice</li> <li>• Applications to hydrosphere, minerals (remote sensing surveys), soils, pollution monitoring</li> <li>• Evaluation: improved coverage and resolution vs cost, calibration, interpretation, remaining uncertainty; justified conclusion</li> </ul> <p><b>Source:</b> R. Genn, Environmental Science A level AQA (Insight &amp; Perspective, 2018), pp. 9, 89-107, 402-407; AQA spec 3.2, 3.7.</p> | 25    |

| Question | Marking guidance                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Marks |
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| 11.2     | <p><b>Mineral resources are non-renewable. Discuss how an understanding of geological processes and the development of new technologies can extend the period for which society can rely on mineral resources.</b></p> <p><b>Levels of response</b></p> <p><b>Level 5 (21-25 marks):</b> Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached.</p> <p><b>Level 4 (16-20 marks):</b> Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified.</p> <p><b>Level 3 (11-15 marks):</b> Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics.</p> <p><b>Level 2 (6-10 marks):</b> Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions.</p> <p><b>Level 1 (1-5 marks):</b> Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate.</p> <p><b>0 marks:</b> No relevant content.</p> <p><b>Indicative content</b> (students are not expected to cover all of these; credit other relevant material)</p> <ul style="list-style-type: none"> <li>• Why non-renewable: formed too slowly; geological concentration processes - hydrothermal, metamorphic, Proterozoic sediments, alluvial, evaporites, secondary enrichment, biological sediments</li> <li>• Understanding processes guides exploration (batholith margins, river gravels, ancient seas) and remote sensing / geophysical techniques</li> <li>• Reserves, resource, stock; Lasky's principle; cut-off ore grade and its drivers</li> <li>• Extraction technologies: mechanisation, bioleaching, phytomining, polymer and bacterial adsorption, seabed nodules</li> <li>• Reducing demand: recycling and its difficulties, Cradle to Cradle, design for disassembly, linear vs circular material flows, dispersal of metals in use</li> <li>• Environmental limits: impacts of mining and their control; energy costs rising with lower grades; conflicts</li> <li>• Judgement: technology can extend supplies substantially but only within energy and environmental constraints; circular economy essential</li> </ul> <p><b>Source:</b> R. Genn, Environmental Science A level AQA (Insight &amp; Perspective, 2018), pp. 130-147, 373-379; AQA spec 3.2.3, 3.6.3-3.6.4.</p> | 25    |